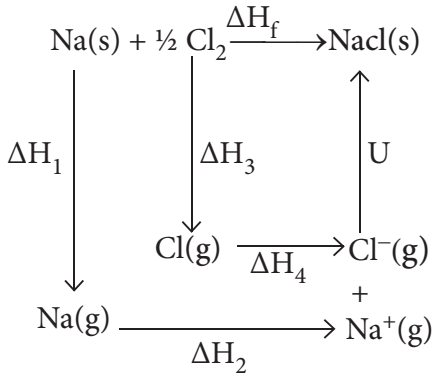
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$\Delta H_f = \Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4 + U$

Let us use the Born - Haber cycle for determining the lattice enthalpy of NaCl as follows:

Since the reaction is carried out with reactants in elemental forms and products in their standard states, at 1 bar, the overall enthalpy change of the reaction is also the enthalpy of formation for NaCl. Also, the formation of NaCl can be considered in 5 steps. The sum of the enthalpy changes of these steps is equal to the enthalpy change for the overall reaction from which the lattice enthalpy of NaCl is calculated.

Let us calculate the lattice energy of sodium chloride using Born-Haber cycle



ΔH_f = heat of formation of sodium chloride = – 411.3 kJ mol^{-1}

ΔH_1 = heat of sublimation of Na(S) = 108.7 kJ mol^{-1}

ΔH_2 = ionisation energy of Na(S) = 495.0 kJ mol^{-1}

ΔH_3 = dissociation energy of Cl_2(S) = 244 kJ mol^{-1}

ΔH_4 = Electron affinity of Cl(S)= – 349.0 kJ mol^{-1}

U = lattice energy of NaCl

$$\Delta H_f = \Delta H_1 + \Delta H_2 + \tfrac{1}{2}\Delta H_3 + \Delta H_4 + U$$

$$\therefore U = \left(\Delta H_f\right) - \left(\Delta H_1 + \Delta H_2 + \tfrac{1}{2}\Delta H_3 + \Delta H_4\right)$$

$\Rightarrow$ U = (–411.3) – (108.7 + 495.0 + 122 – 349)

U = (–411.3) – (376.7)

$\therefore$ U = – 788 kJ mol^{-1}

This negative sign in lattice energy indicates that the energy is released when sodium is formed from its constituent gaseous ions Na^+ and Cl^-

Evaluate Yourself - 4

When a mole of magnesium bromide is prepared from 1 mole of magnesium and 1 mole of liquid bromine, 524 kJ of energy is released.

The heat of sublimation of Mg metal is 148 kJ mol^{-1}. The heat of dissociation of bromine gas into atoms is 193 kJ mol^{-1}. The heat of vapourisation of liquid bromine is 31 kJ mol^{-1}. The ionisation energy of magnesium is 2187 kJ mol^{-1} and the electron affinity of bromine is – 662 kJ mol^{-1}. Calculate the lattice energy of magnesium bromide.

7.10. Second Law of thermodynamics:

Need for the second law of thermodynamics:

We know from the first law of thermodynamics, the energy of the universe is conserved. Let us consider the following processes: